REP-006 · PLAN-004 done: the online "stuck on Connecting..." bug is found and fixed

Game	Fruit Market
Report date	2026-10-26 15:25 (Asia/Kuala_Lumpur)
Package	PLAN-004, approved by the owner 2026-10-26: "Approve PLAN-004 only, including its listed tests and documentation"
Build	`main @ d7e5d24` (fix in `9d9e911`)
Live game	**Unchanged** (1.1.0). The fix is on the preview; players get it only after "PREPARE RELEASE" and your Deploy
Delivery status	Verified (R1–R6 PASS)

The bug, in plain words

After a player reloaded the page or rejoined an online match, the game sometimes stayed on **"Connecting to room ..."**, even though the match was there and the server had already sent it.

Why: the screen started listening for room updates a moment *after* it first appeared. The connection was already open by then. If the server answered in that gap, the answer was missed, and the screen waited for a next message that might never come (for example, when everyone was waiting on that player).

The gap is tiny on a fast computer, so it usually worked. On a slow phone, or when the app is in the background (browsers pause drawing for hidden tabs), the gap is long and the bug happens often. The game screen had the same weakness for updates arriving just as it opened.

Fix: two small changes in the app (`src/ui/Online.tsx`, `src/ui/game/useGameController.ts`). Right after starting to listen, each screen checks what has already arrived.

- No server change, no message-format change, no rules or save change.
- Old and new app versions stay compatible.

Evidence

Check	Result
Normal timing, old code (local)	270 cycles (reloads, network drops, quick matches; normal and 4× CPU slowdown), 0 failed. The bug is too rare to see this way
Background-tab mode, **old code**	**20 of 20 reloads failed**; every one stuck on "Connecting to room" with the full game already received (`room v2 status=playing`)
Background-tab mode, **fixed code** (local)	65 cycles, 0 failed

Check	Result
Background tab plus 4× slowdown, **fixed code** (local)	100 cycles, 0 failed (reload p95 0.59 s, drop p95 0.63 s, quick match p95 1.3 s)
Same on a CI machine (where the original failures happened)	110 cycles, 0 failed (Reconnect stress run 40283316566)

Acceptance results

#	Criterion	Result	Evidence
R1	Stress test with per-cycle timings	PASS	`npm run test:reconnect` ; modes: slowdown, background tab, preview target
R2	Root cause named with evidence	PASS	Lost first update after (re)joining; see the old-code evidence above
R3	Before rate recorded; at least 100 slowed cycles with 0 failures after	PASS	20/20 failed before; 100/100 passed after (plus 175 more)
R4	Forced failure leaves evidence	PASS	Page text, room messages, console, both screenshots and the server log were saved
R5	Regression test fails before, passes after	PASS	New suite step "rejoin from a background tab"
R6	3 consecutive green CI runs; leak check 0; unit tests green; no rule or save changes	PASS	CI runs 7 (push), 8 and 9 (manual) on `d7e5d24` : all green, each with leaks 0 and 63 unit tests; no rules or save files changed

Side observations (no action taken)

- **The local test server never finishes closing a socket when the page closes it.** Cloudflare's docs say production replies automatically for this project's settings, and the real app never waits on a socket it closed itself, so no player impact was found. If it ever matters, the documented safe fix is one line in the server's close handler.
- **Playwright's offline mode doesn't break open connections**, so the stress test cuts them with a small local relay instead. A "silent" dead connection (no signal at all) isn't simulated yet. That would be part of Package C.

What changed

Added

- `npm run test:reconnect` and a manual *Reconnect stress* workflow (local or preview, optional slowdown and background-tab mode).
- Failure evidence for all online browser tests.
- The regression step in the browser suite.

Fixed (player-facing, on the preview, not yet live)

- After reloading or rejoining an online match, the game could stay stuck on "Connecting to room...", especially on slower phones or when the app was in the background.

Risks and costs

- RM0.
- The browser suite is about 1 minute longer (the regression step).
- The fix affects all online screens. Mitigation: the full suite, including the secret-price leak check, stayed green, and 545 stress cycles ran in total.
- Remaining NOT RUN: a real phone in a real background tab. Your preview playtest covers that (see below).

How to try it (preview)

1. Open https://fruit-market-staging.murshidi-hashim.workers.dev on two phones (or one phone and a computer).
2. Start an online game (Play a friend → Create room → open the link on the other device).
3. During the match, on one phone: switch to another app for 10 seconds, come back, then pull down to reload.
4. Expected: the match comes back within a few seconds both times, never stuck on "Connecting to room...".